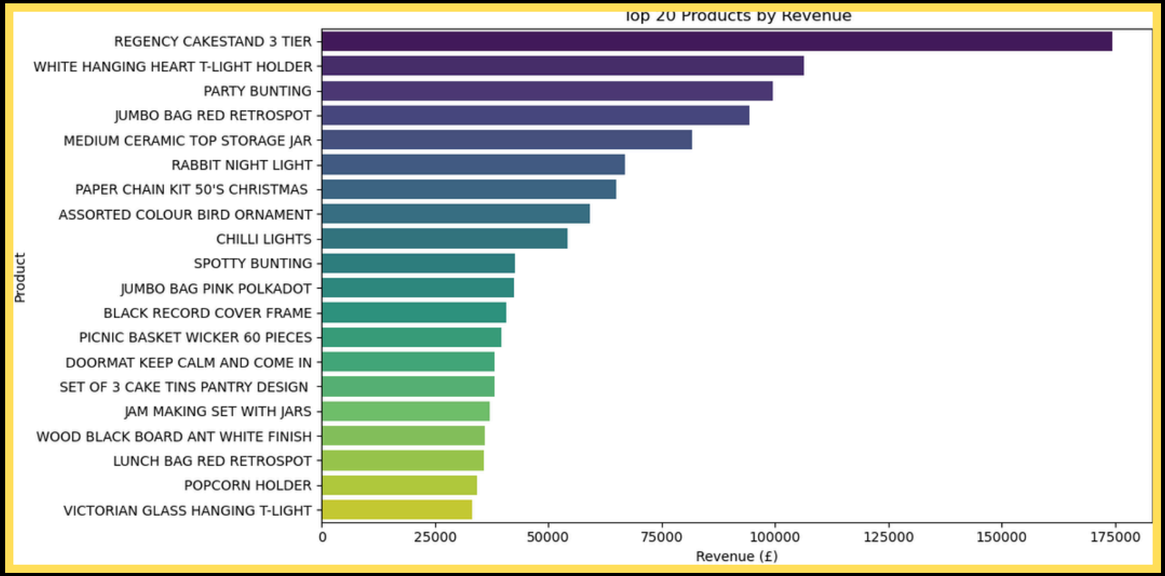
UK Online Retail Analytics

Goal:

To analyze historical data from a **UK online retail store**, identify key products, trends over time and geography, as well as data anomalies, in order to provide business **insights** for *optimizing sales*, *product assortment*, and *marketing strategies*.

Top 20 Products by Revenue



Insights:

Total sales are heavily concentrated in a small group of party and decor items, with the top 5 products significantly outperforming the rest of the list.

Recommendations:

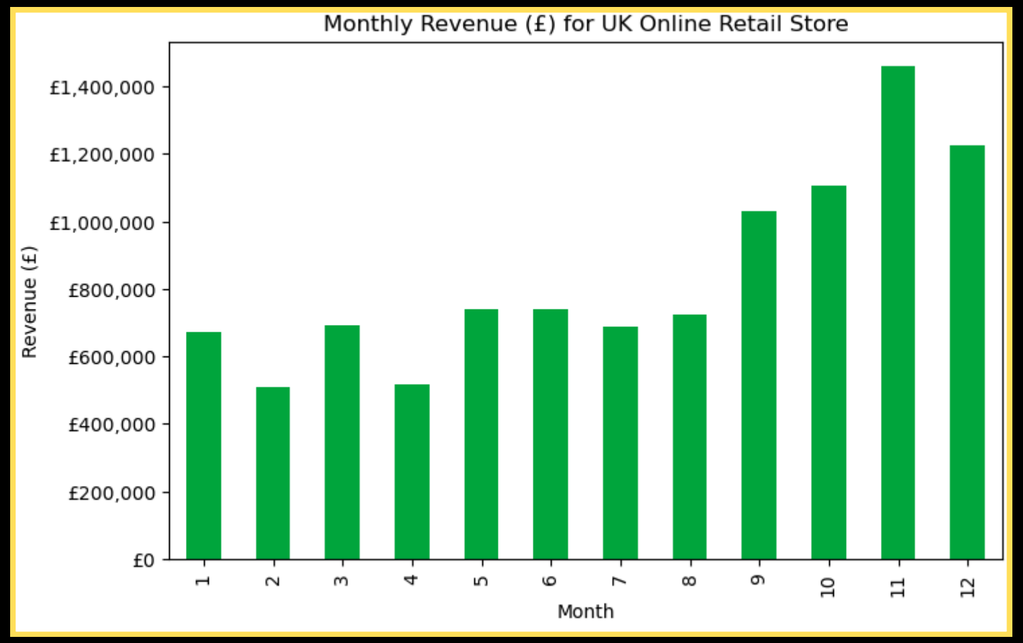
Focus budget: direct marketing efforts toward the top 5 products to maximize return on investment.

Maintain stock: prevent inventory shortages for bestsellers, especially the cakestands, to avoid lost revenue.

Create bundles: combine top-performing items with lower-ranked products to increase the average order value.

Sales by Time

Goal: detect seasonal peaks, optimize marketing campaigns and operations



Insight:

The data reveals a steady upward trend with a massive surge in the fourth quarter. Revenue peaks in **November**, which is nearly triple the earnings of the lowest month, **February**. This highlights a powerful seasonal sales cycle, likely driven by holiday shopping and year-end promotions.

Recommendations:

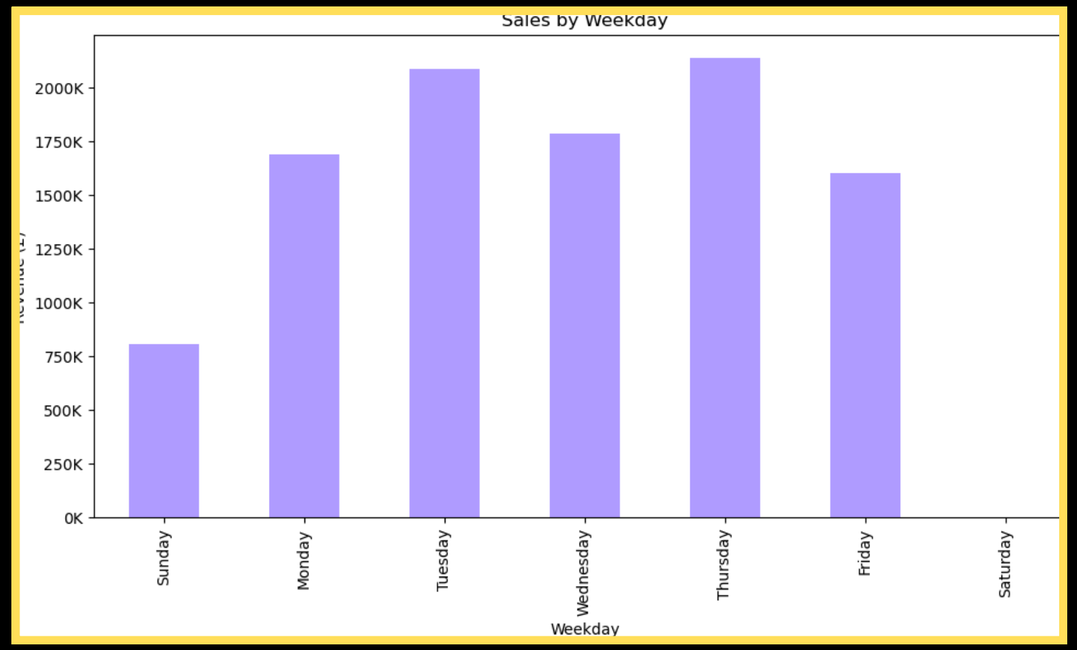
Capitalize on Q4: Maximize inventory and ad spend from **September** to **November** to ride the peak trend.

Stimulate Low Periods: Launch aggressive promotions in **February** and **April** to stabilize cash flow.

Retain Peak Buyers: Offer post-holiday incentives to convert year-end shoppers into **Q1** customers.

Scale Early: Start ramping up operations in **August** to be fully ready for the **September** growth surge.

Sales by Weekday



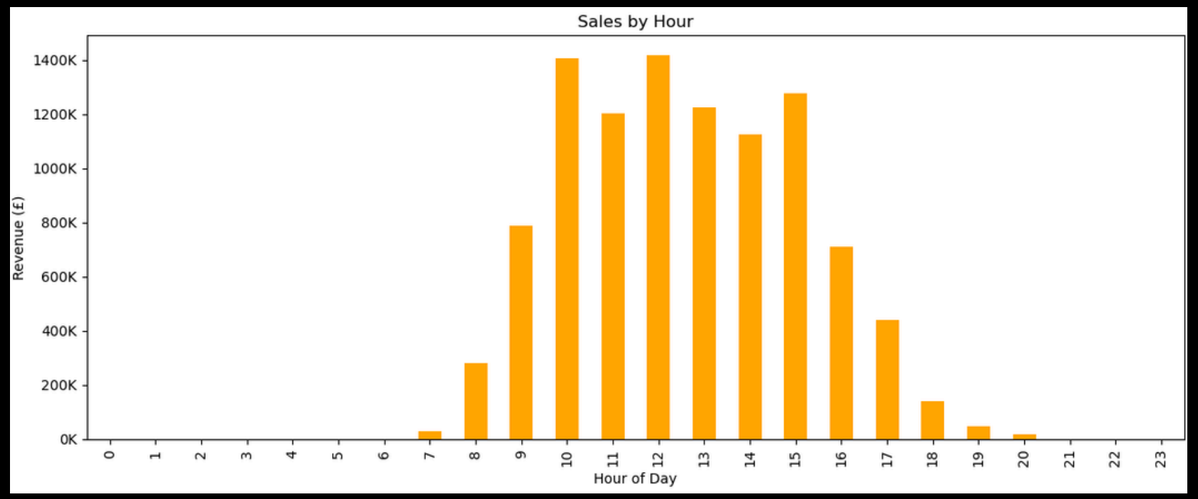
Insight:

Sales peak on **Thursday** and **Tuesday**, while **Saturday** shows zero sales. **Sunday** is the weakest active day.

Recomendations:

Schedule campaigns: plan major marketing activities and newsletters for Tuesday and Thursday to capture peak buying interest.
Boost weekends: introduce exclusive "weekend-only" deals to drive sales on Saturday and Sunday.
Scale mid-week resources: increase staffing and logistics support on high-volume days to ensure fast order processing.
Audit Saturday gap: investigate why Saturday sales are at zero to fix potential technical issues or operational closures.

Sales by hours



Insight:

Peak shopping activity occurs between **10:00** and **15:00**. Revenue spikes at **10** and **12**, followed by a sharp decline. Evening and nighttime hours show almost zero activity.

Recomendations:

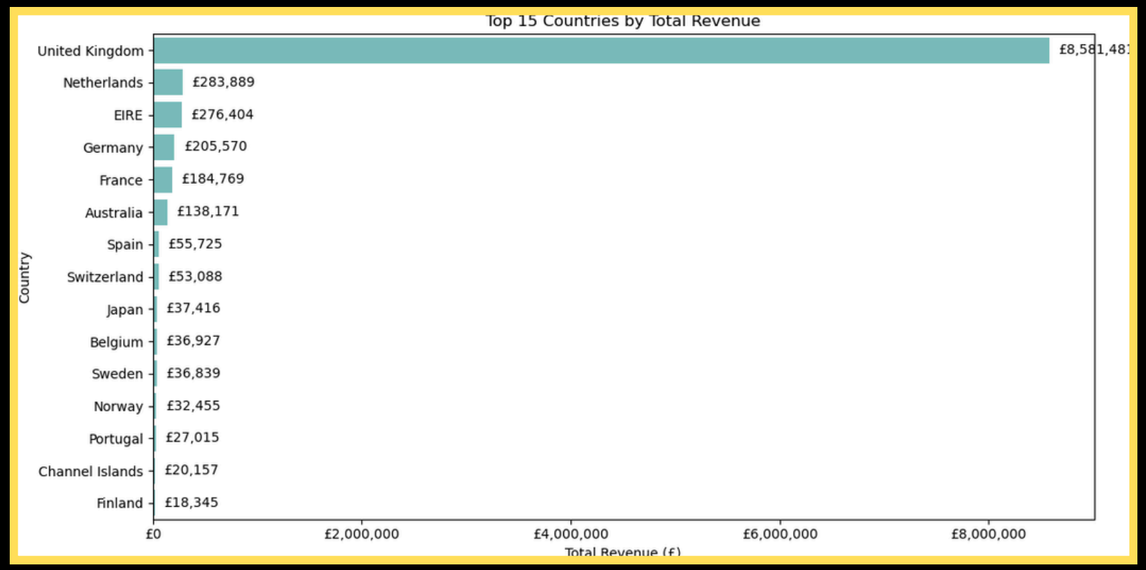
Optimize ad delivery: focus the majority of your daily advertising budget between **9:00** and **15:00** to catch customers during their most active buying window

Time newsletters: send email or SMS campaigns at **9:30** to stay at the top of customers inboxes exactly when the daily peak begins.

Align customer support: schedule the maximum number of support staff for the **10:00–16:00** shift to handle the high volume of inquiries and orders.

Sales by country

Goal: identify key markets and growth opportunities



Insight:

The **United Kingdom** is the dominant market. This is dozens of times higher than the next closest markets (Netherlands and EIRE). The business is heavily dependent on a single region, while international sales remain in the early stages of growth.

Recomendations:

Maintain UK leadership: prioritize resources to protect the UK market share, as it serves as the core financial foundation.
Scale EU presence: increase marketing investment in the **Netherlands**, **Germany**, and **France**, as they show the highest international potential.
Localize marketing: adapt product descriptions and customer support to the local languages of top European markets to boost conversion rates.
Optimize global shipping: review delivery costs for lower-revenue countries to make the offer more competitive.
Analyze Australian potential: investigate the drivers behind high sales in this distant region and consider expanding operations there.

Customer Segmentation

Goal: Identify key customer segments, quantify their revenue, and guide targeted marketing and retention

Insight:

1. Loyal and Frequent (708 customers):
- Contribute most of the store's revenue.
- Purchase often with consistent average order value.

2. High Revenue / VIP (159 customers):
- Purchase rarely but very high-value items.
- Their behavior heavily influences total revenue.

3. New / Inactive (2627 customers):
- Recently bought or haven't purchased in a while.
- Large potential for repeat purchases if engaged.

4. Lost (841 customers):
- No longer purchasing.
- Represents lost sales; reasons should be analyzed and addressed.

Recomendations:

1. For Loyal and Frequent:
- Implement loyalty programs or bonuses to encourage more frequent purchases.
- Offer personalized promotions on products they already like.

2. For High Revenue / VIP:
- Provide premium offers and campaigns for high-value products.
- Offer personalized service or VIP shipping.

3. For New / Inactive:
- Send reminders via email/SMS; offer discounts for the first repeat purchase.
- Segment further by last purchase date for precise targeting.

4. For Lost:
- Analyze why they stopped purchasing